Professional Issues in IT

STRUCTURE AND MANAGEMENT OF ORGANIZATIONS

Structure and Management of Organizations

How Organizations are structured?

How can you suggest alternative possible structures and identify their advantages and disadvantages.

Introduction

Definition: An organization is a group of people working together in a formal way.

Explanation: The work that has to be done is shared between these people and there are rules about who does what.

How the work is shared and how tasks and people are grouped together – the structure of the organization – will vary very much from organization to organization. It is surprising, however, that organizational structures have much more in common than might be expected.

ORGANIZATIONAL MODELS

1. The bureaucratic model
2. The organic model
3. Matrix management

The bureaucratic model

- Organizational theory is the study of how organizations are structured and how they work.(19th century)
- The founders of the theory were sociologists like Max Weber and Mary Parker Follett, and practical business people like Henri Fayol and Lyndall Urwick.
- In a modified form, this model still describes the organizational structures to be found in most large, and many smaller, organizations.

The bureaucratic model characteristics

The ideal bureaucratic organization was thought to have the following characteristics:

1. All tasks are split up into specialized jobs, in which jobholders become expert; management can thereby hold the jobholders responsible for the effective performance of their duties.
2. The performance of each task is governed by precise rules. This means that there should be no variation in the way tasks are carried out and therefore no problems with the co-ordination of different tasks.
3. Each individual (and hence each unit) in the organization is accountable to one and only one manager.

Characteristics (cont…)

4. In order to ensure that personalities and personal relationships do not interfere with the organization's performance, employees are required to relate both to other employees and to clients in an impersonal and formal manner.

5. Recruitment is based on qualifications and employees are protected against arbitrary dismissal. Promotion is based on seniority and achievement. Life-time employment is envisaged.

The vertical world of hierarchical bureaucracy



- Roles
- Rules
- Plans
- Reports

The organic model

- The best known alternative model is the *organic* model
- Rensis Lickert expresses the basic assumption of the model in the following (rather verbose) terms:
- **"An organization will be effective to the extent that its structure is such as to ensure a maximum probability that in all interactions and in relationships within the organization, each member, in the light of his background, values, desires, and expectations, will view the experience as supportive and one which builds a sense of personal worth and importance."**

Characteristics of the Organic Model

- Decentralized Decision-Making
- Low Specialization and Formalization
- Collaboration and Flexibility
- Emphasis on Expertise

Organic model

This view underlies the organizational structure of most small professional companies – software houses, advertising agencies, even solicitors' and GPs' practices

it is also common in academic institutions, both schools and universities. The view is not necessarily consciously articulated – nor is this view and the adoption of the structures it suggests sufficient to achieve effectiveness!

Proponents of the bureaucratic model claim that it is universally applicable.

Proponents of the organic model make similar claims.

It says little for common sense that those who hold the obvious view that each has its appropriate place should be christened adherents of the contingency school of organizational design.

Matrix management

- It is an essential feature of the bureaucratic model that every individual and every unit in the organization is responsible to only one manager.
- This is not realistic in the context of project-based, high-technology companies.
- It accepts that individuals may be responsible to more than one manager and requires rules that will enable possible conflicts to be resolved.
- Some organizations and some management consultants have tried to formulate the matrix management model much more formally. The results are not encouraging.
- A specialist in high-speed communications working for a systems integrator may well find themselves working on two or three projects simultaneously, as well as having a more general responsibility for maintaining the company's expertise in the area.
- In the past 30 years or so the idea of matrix management has become fashionable as a way of addressing such situations.

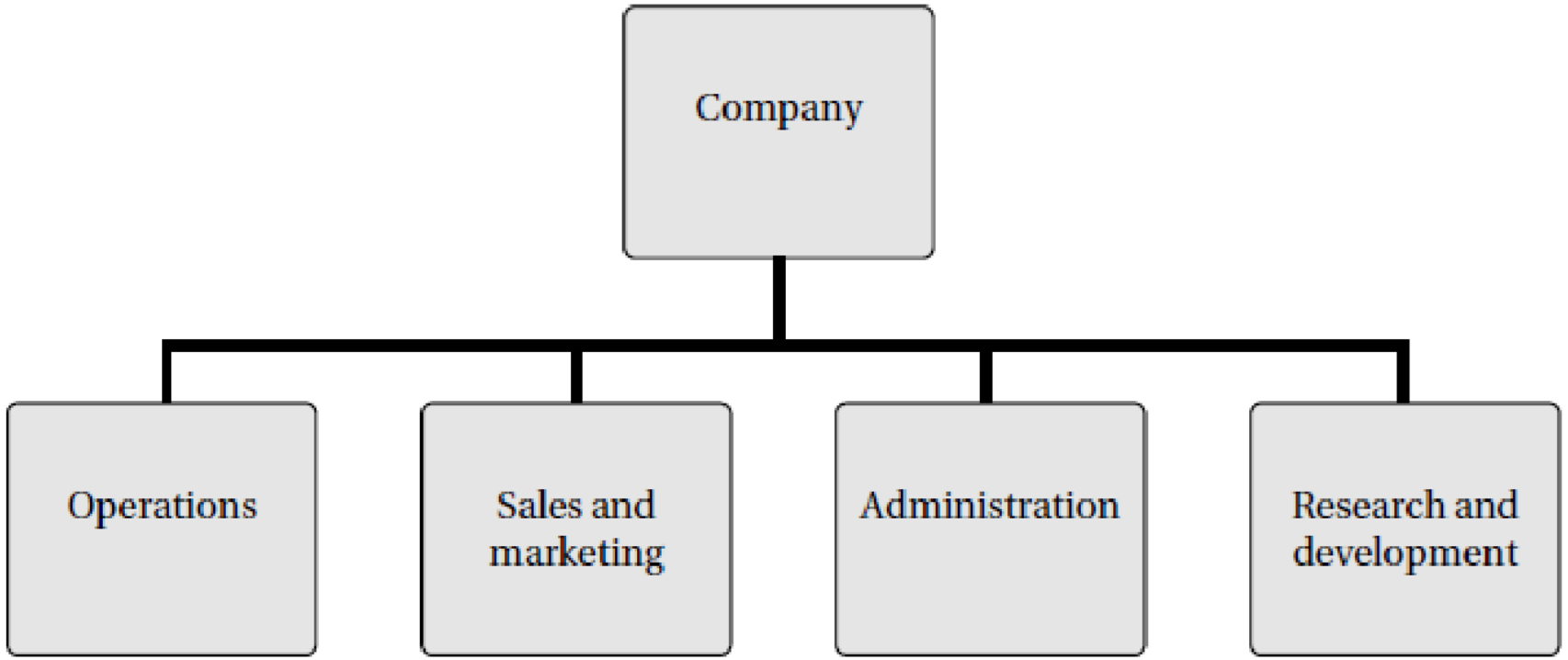
STRUCTURING PRINCIPLES

- Structure by function
- Structure by geography
- Product line structure
- Structure by market sector
- Structure by technology
- Operational structure

Structure by function

- There are the activities that are the primary purpose of the organization. These activities are known as *operations*.
- All organizations have to pay their bills and pay their employees. They need to ensure that the buildings they use are cleaned regularly. If they charge for their services, they may need to send out bills and ensure that these are paid. They will probably need to hire new employees from time to time. These activities are generally known as *administration*.
- Many organizations will need to publicize their services or their products and try to persuade people to use them or buy them. In the business world these activities are usually known as *sales and marketing*.
- Finally, many organizations need to be continually developing new products or services, or developing new ways to deliver them. These activities are known as research and development.

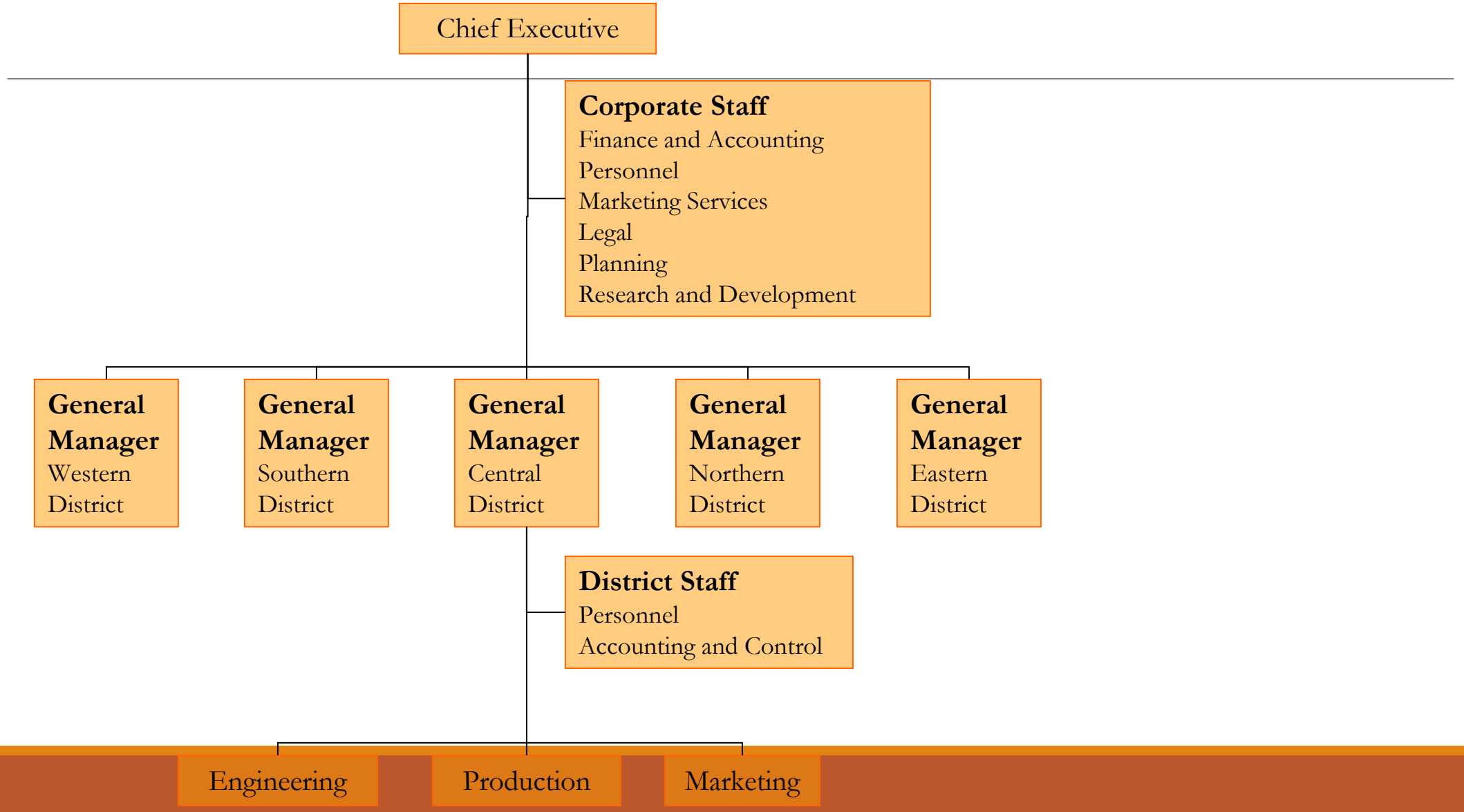
Structure by function



Structure by geography

- In many cases it makes sense to group activities together on a geographical basis. Multinational companies, that is, companies that operate in a number of different countries, are usually forced to have some geographical elements in their structure.
- The subsidiaries are subject to the laws of the countries in which they are registered, in particular, the laws regarding employment, accounting and taxation.
- Within a single country, geographical factors have become less important as a result of the development of modern communications and, as a result, geographical structures have been replaced by structures based on other factors.

Geographic Organizational Structure



Geographic Organizational Structures:
Strategic Advantages/Disadvantages

Strategic Advantages

Allows tailoring of strategy to needs of each geographic market

Improves coordination within the market

Takes advantage of economies of local operations

Strategic Disadvantages

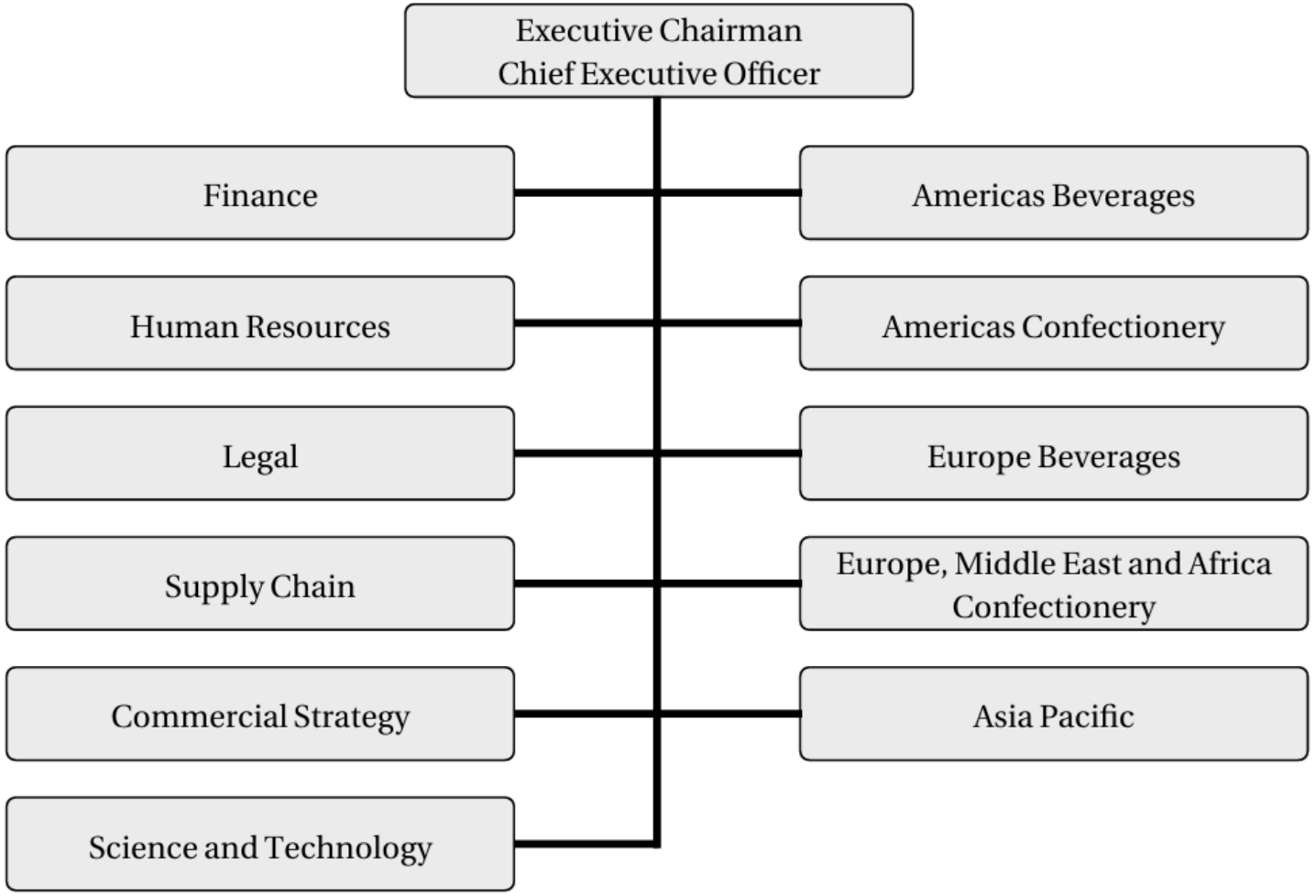
Greater difficulty in maintaining consistent company image across areas

Can result in duplication of staff services at headquarters and district levels, making a relative-cost disadvantage

Product line structure

- A product line structure is a structure that is based around the different types of product that an organization produces.
- For example, where a motor vehicle manufacturer organizes around types of vehicle.
- Companies that produce and market a substantial piece of software for corporate customers – a multi-user accounting package
- For example – often organize themselves into three main operational divisions: development and maintenance of the software, consultancy, and training. This should be regarded as a product line structure since the three types of activity, providing software, giving advice to companies in how to use it, and providing training for customer staff, can be considered to be different services that the company provides and they are typically provided by different teams of people.

Product line structure



Structure by market sector

- Structure by market sector means structure based on the different market sectors to which its customers or prospective customers belong.
- From the sales and marketing point of view it has the great advantage that each division can fairly readily identify its potential customers, and its staff, both sales and technical, are likely to be familiar with customers' problems and to speak a language that the customer understands.
- Example :

Different branches of BATA

Automobiles

Structure by market sector Disadvantages

There are two dangers with this approach.

- There is the risk one division may be unaware of technological expertise that exists in another division. This may lead to inefficient use of resources through unnecessarily hiring additional specialists or employing consultants, or, worse, to failing to learn from mistakes that have been made by other parts of the company.
- The second danger with a structure based on market sector is that, by continuing to concentrate on its traditional areas even when these markets are becoming saturated, the company will miss new opportunities and will stagnate.

Structure by technology

A technology-based structure was once a favorite model for software companies.

A company might have divisions specializing in artificial intelligence, communications, web-based systems, databases, and real-time systems.

Structure by technology

- There are several problems with type of structure: it usually requires several different technologies to meet a customer's needs;
- there are many applications that cannot be said to require specific technologies;
- there are many competent software engineers whose expertise runs across a number of technologies;
- it is difficult, if not impossible, for sales and marketing staff to predict which potential clients will need which technology. It is particularly serious and companies that are primarily structured by technology have serious problems finding their clients.
- They are not sufficiently 'customer-focused' – they concentrate on selling the technologies that they have rather than finding out what the customer needs.

Operational structure

- The actual operations of a company may be organized on a *project* basis or on a *production* basis, although the line separating the two may be vague.
- Project-based activity is not restricted to operations. Most research and development is organized on a project basis and such administrative activities as introducing a new accounting system or transferring a company's head office are also to be regarded as projects, in that they last for a fixed length of time, after which they should be complete.
- Projects last a comparatively long time but the team carrying out the work only stays together for the length of the project. Production activities are comparatively short, but the team carrying them out stays in existence indefinitely.
- The central data processing operations of a com pany are organized on a production basis. There is a schedule of programs payroll, accounts payable, accounts receivable, and so on – that have to be run regularly on specific dates

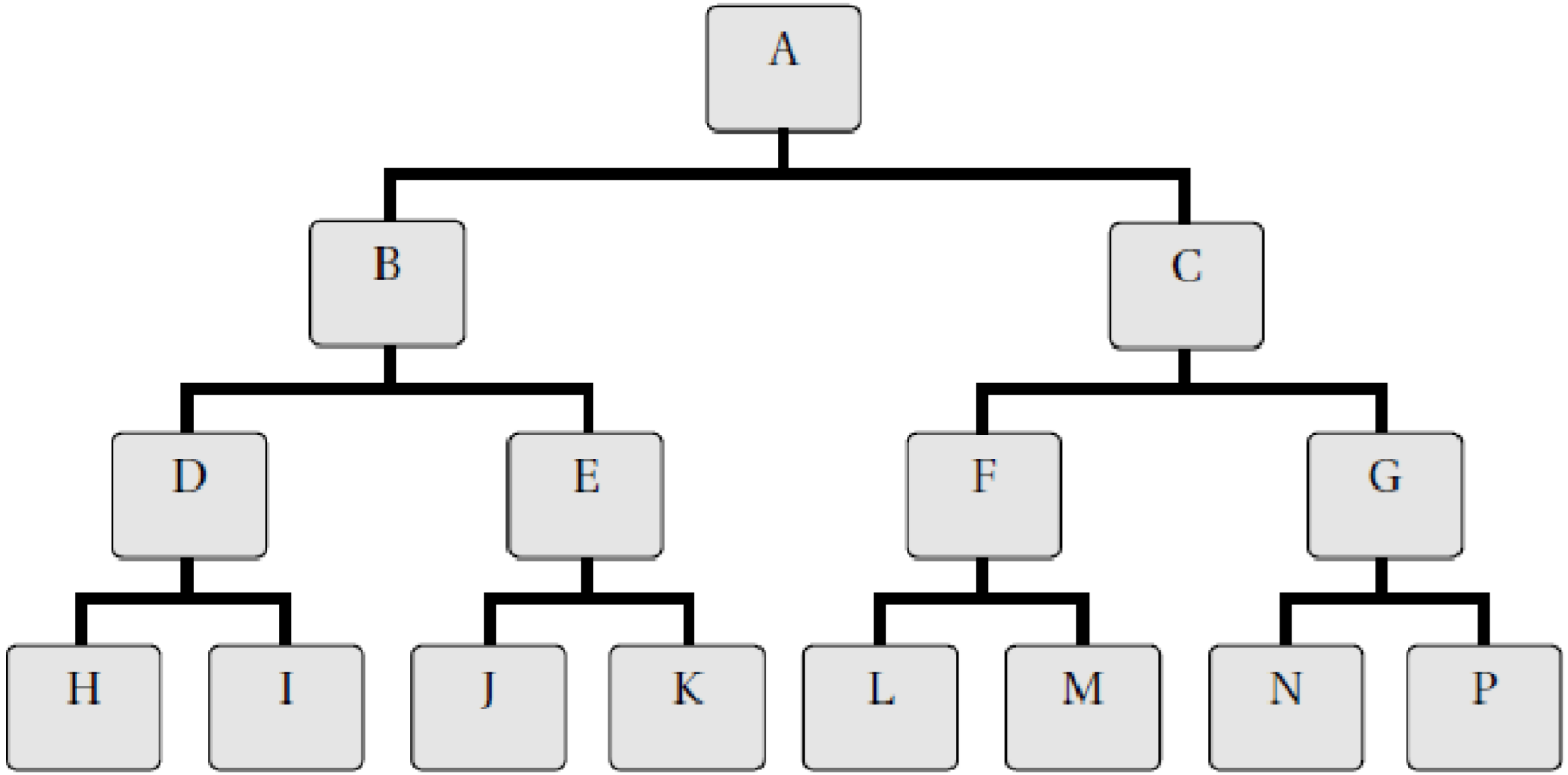
Discussion

- Discuss the types of industries where production-based work is more common versus industries that rely heavily on project-based work. Why is each structure more suitable for these industries?
- In a rapidly changing business environment, how does the flexibility of project-based teams give organizations a competitive advantage?

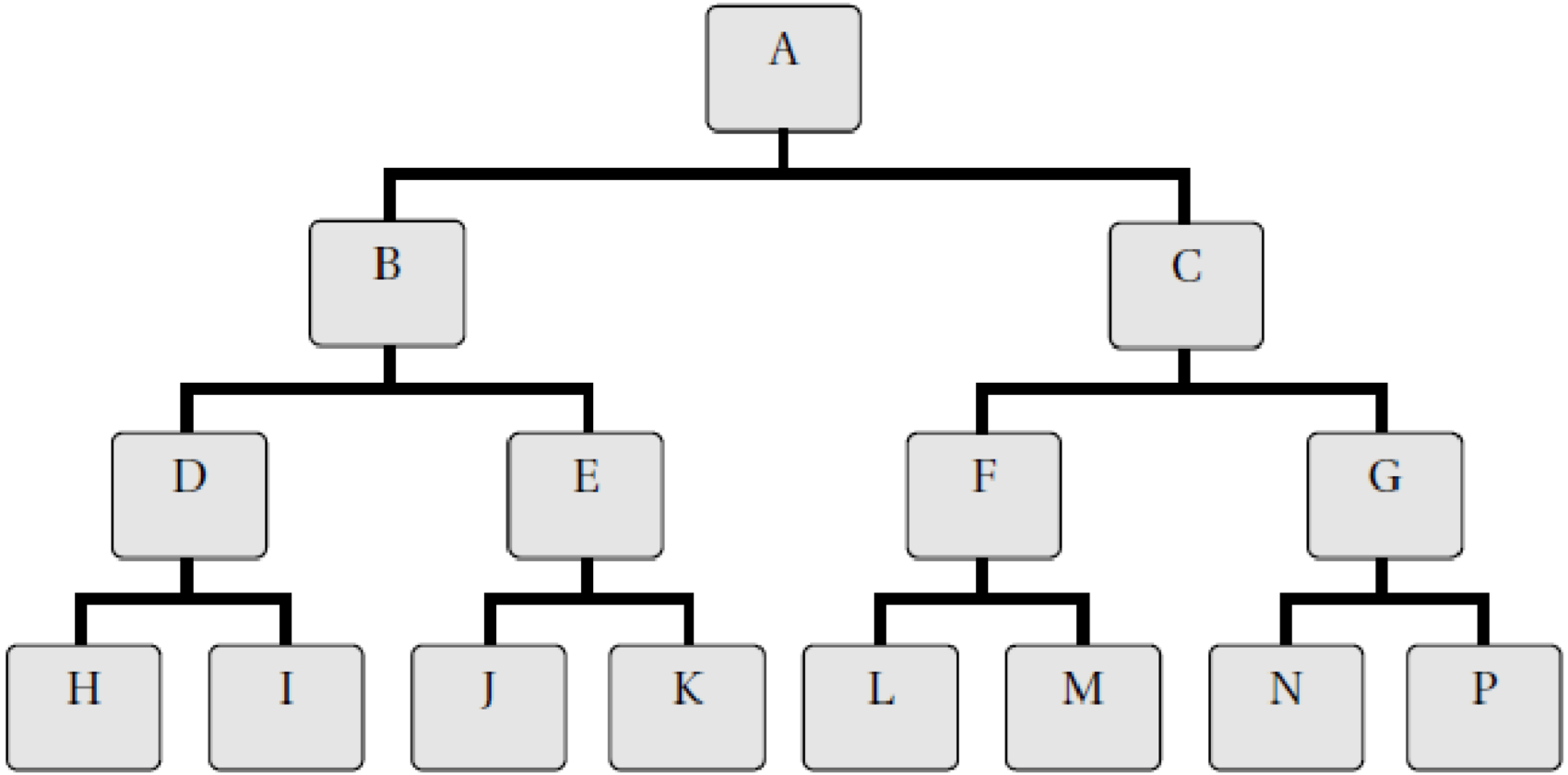
DEPTH OF STRUCTURE

The depth of an organizational structure is the number of layers in the structure – or, more precisely, the maximum number of layers, since not all parts of the structure will have the same number of layers. Organizational structures are often described as flat or, in contrast, deep or tall, according to whether the depth is small or large. For a given number of people, the depth of the structure will obviously depend on the number of people reporting directly to each manager; this is sometimes known as the manager's *span of control*.

Fifteen people organized into a four-level structure



Fifteen people organized into a three-level structure



CENTRALIZATION

- Organizations may be centralized or decentralized. In a centralized company, as much power as possible is kept at the top of the company, with delegation only when essential. In a decentralized company, as much power and control as possible is delegated to the lowest level.

- Drawing the correct balance between centralization and decentralization is important but difficult. Decentralization is commonly found in high-technology companies, where there is plenty of talent at lower levels. Centralization is commoner in large manufacturing companies and other long-established organizations.

- The ideal might be described as flexible centralization, in which rules and practices are laid down centrally but it is accepted that reasonable arguments for modifying them in specific cases will be readily accepted.

DISCUSSION

- Pontevedro Software produces a range of packages for project management, accounting, and cost estimation in the construction industry. It is currently developing a new package for computer aided drafting that will interface with the cost estimation package. It currently employs some 200 professional staff. The packages are sold in the UK, North America and Australia but the company is keen to expand into Asia and into mainland Europe. The organizational structure of the company is more or less the same as when it was a company with 20 staff, operating only in the UK, and the directors have decided that a fundamental reorganization is necessary.
- Discuss the main decisions that have to be taken before the new organization can be set up

SETTING UP A STRUCTURE IN PRACTICE

In most cases, an organization of any size will have a structure that includes elements of several of the different types of structure.

A combination of different structures would help when adopted.

An organizational structure for a bespoke software house

